Thankfully, it was quite easy to do. A few word changes in critical paragraphs softened the edge of our claims without compromising the technical content. Within an hour, we resubmitted the paper, and agreed not to talk about it at all during the drive to Morrison's house.

By early the next afternoon it was evident that the response to our paper was enthusiastic. Among the many e-mail responses was one from Plesser, who gave us one of the highest compliments one physicist can give another by declaring, "I wish that I had thought of that!" Notwithstanding our fears the previous night, we had convinced the string theory community that not only can the fabric of space undergo the mild tears discovered earlier (Chapter 11), but that far more drastic rips, roughly illustrated by Figure 13.3, can occur as well.

Returning to Black Holes and Elementary Particles

What does this have to do with black holes and elementary particles? A lot. To see this, we must ask ourselves the same question we posed in Chapter 11. What are the observable physical consequences of such tears in the fabric of space? For flop transitions, as we have seen, the surprising answer to this question was that not much happens at all. For conifold transitions—the technical name for the drastic space-tearing transitions we had now found—there is, once again, no physical catastrophe (as there would be in conventional general relativity), but there are more pronounced observable consequences.

Two related notions underlie these observable consequences; we will explain each in turn. First, as we have discussed, Strominger's initial breakthrough was his realization that a three-dimensional sphere inside a Calabi-Yau space can collapse without an ensuing disaster, because a three-brane wrapped around it provides a perfect protective shield. But what does such a wrapped-brane configuration look like? The answer comes from earlier work of Horowitz and Strominger, which showed that to persons such as ourselves who are directly cognizant only of the three extended spatial dimensions, the three-brane "smeared" around the three-dimensional sphere will set up a gravitational field that looks like that of a black hole.2 This is not obvious and becomes clear only from a detailed study of the equations governing the branes. Again, it's hard to draw such higher-dimensional configurations accurately on a page, but Figure 13.4 conveys the rough